Lab Report: Time Series Econometrics

ANANDARAJAH Rahul

Written on March 20, 2026

The lab sessions were carried out using the `R` language. The code can be found in the appendix of this report.

1. Lab 1: Chickenpox Data

`varicelle.dat` is a file containing the number of chickenpox cases recorded in New York from January 1931 to June 1972. Let us plot this time series:

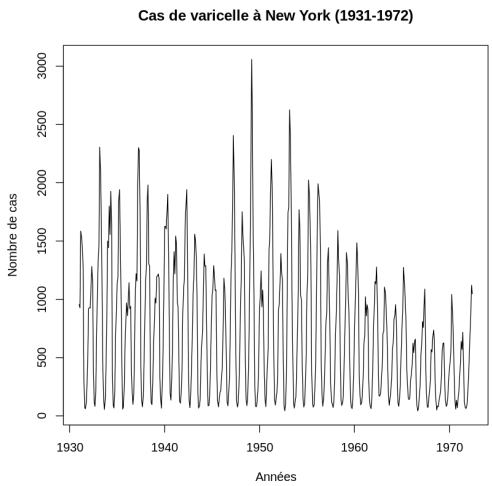

Figure 1: Number of chickenpox cases in New York from January 1931 to June 1972.

At first glance, a downward trend in the total number of cases is observed. However, closer inspection reveals a periodicity, with peaks in March and troughs in August each year.

The average monthly number of cases is 732.41. Let us plot the first 25 autocorrelations:

Autocorrelation is a measure of the linear dependence between a variable and a lagged version of itself. The following observations can be drawn:

- At lag 0, the autocorrelation equals 1, as expected, since a variable is perfectly correlated with itself.
- For other lags, positive and negative autocorrelation spikes appear, indicating a temporal correlation structure in the chickenpox case series.
- These positive and negative spikes suggest a periodicity in the evolution of chickenpox cases, marked by cycles of infection and remission.
- The decreasing envelope of these spikes reveals that the influence of past events gradually diminishes as the lag increases.

To confirm these observations, let us plot the monthly evolution of the number of cases and the annual total.

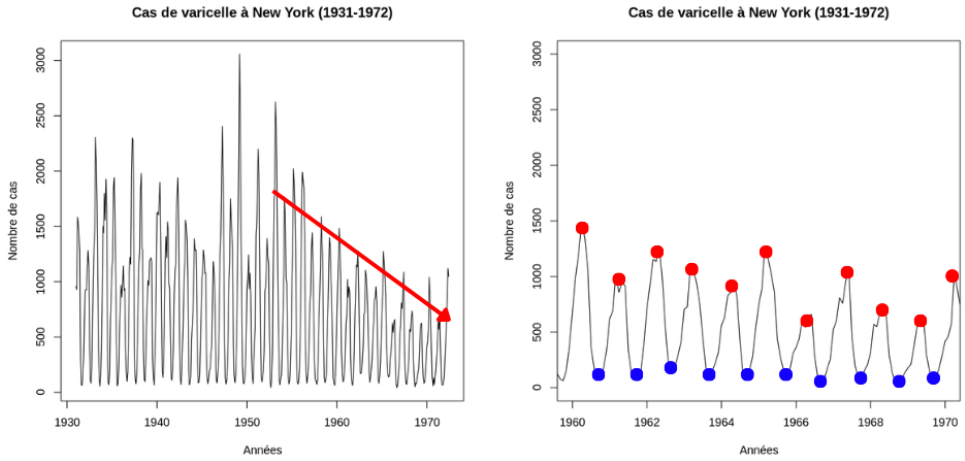

Figure 2: On the left, the convex envelope of the function shows a decrease over time. On the right, the number of chickenpox cases in New York between 1960 and 1970 shows red peaks in March and blue troughs in August.

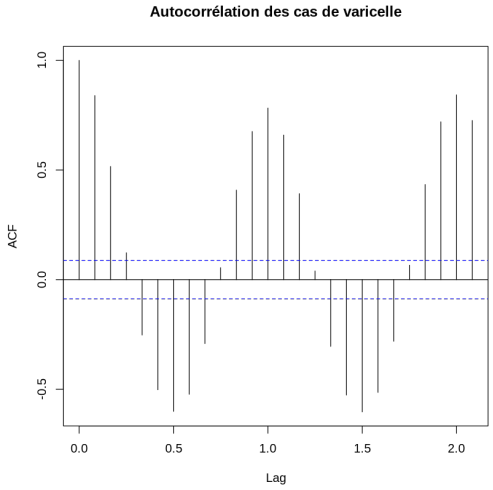

Figure 3: The first 25 autocorrelations of the time series.

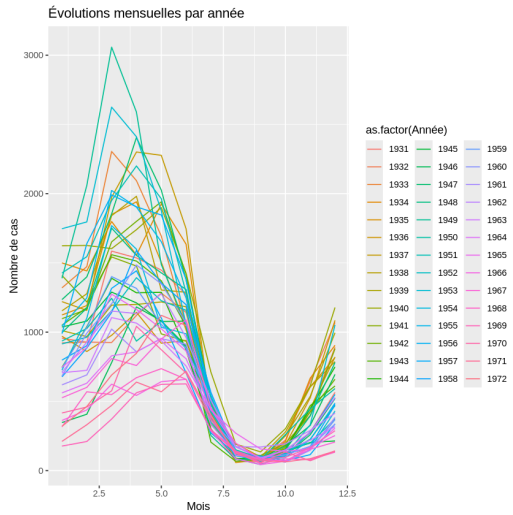

Figure 4: Monthly evolution of the number of chickenpox cases from 1931 to 1972.

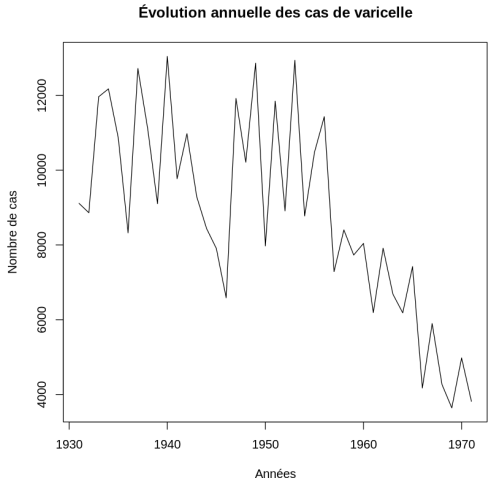

Figure 5: Annual number of chickenpox cases from 1931 to 1972.

The same temporal trends are observed: an increase up to a peak in March, followed by a decrease reaching a trough in August. The blue and yellow curves, corresponding to older data, show higher peaks, while the more recent curves, represented in pink, show a noticeably lower peak. These observations confirm the idea of a gradual decrease in the annual number of cases over time, a hypothesis also supported by Figure 5.

2. Lab 2: Smoothing and Forecasting of Simulated Data

We will simulate the following three time series, with $\varepsilon_t \sim \mathcal{N}(0,1)$ independent and identically distributed:

$$\begin{cases} X_1(t) &= \varepsilon_t \\ X_2(t) &= 0.2t + 2\varepsilon_t \\ X_3(t) &= 0.5t + \varepsilon_t + 3\cos\left(t\frac{\pi}{6}\right) \end{cases} \tag{1}$$

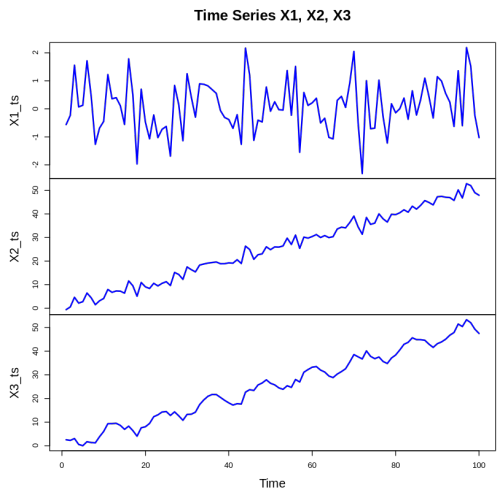

Figure 6: Time series: (1) Gaussian random noise, (2) affine function with Gaussian noise, (3) noisy trigonometric function with an increasing affine trend.

We will smooth the time series using simple and double exponential smoothing with the following values of α: 0.1, 0.3, 0.5, 0.7, and 0.9. We will also apply Holt-Winters exponential smoothing.

2.1. Simple Exponential Smoothing

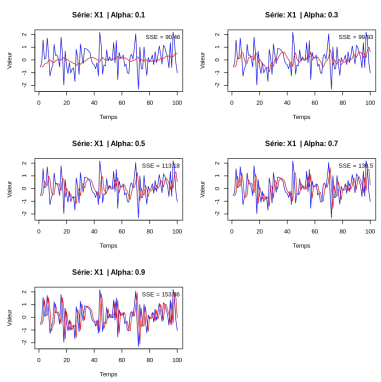

Figure 7: Simple exponential smoothing of X_1 for different values of α.

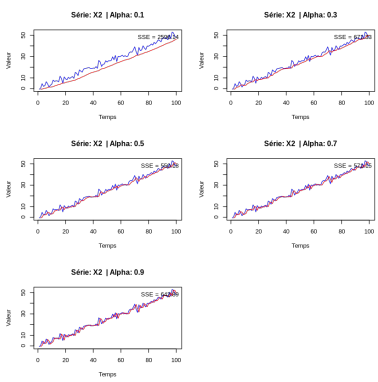

Figure 8: Simple exponential smoothing of X_2 for different values of α.

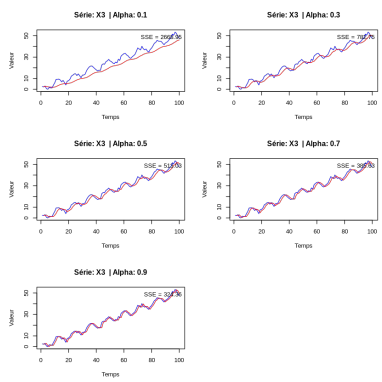

Figure 9: Simple exponential smoothing of X_3 for different values of α.

Visually, it appears that for simple exponential smoothing, the optimal value $\alpha = 0.9$ provides the best approximation of the real curves for all three time series. Let us minimize the sum of squared errors (SSE) to determine the optimal values of α for each series.

α	X_1	X_2	X_3
0.1	90.46297	2596.1359	2668.9533
0.3	99.93229	671.3288	782.7496
0.5	113.17556	550.1759	513.0257
0.7	130.50303	571.1457	385.6319
0.9	153.35624	642.8934	324.3610

Table 1: SSE for all series and all values of α. When the optimization criterion is SSE, $\alpha = 0.1$ is optimal for X_1, $\alpha = 0.5$ is optimal for X_2, and $\alpha = 0.9$ is optimal for X_3.

The obtained values of α differ considerably from those expected. This is because, for large values of α, the smoothing captures the shape of the curve more precisely, but the curve is then shifted, which generates a significant error.

2.2. Double Exponential Smoothing

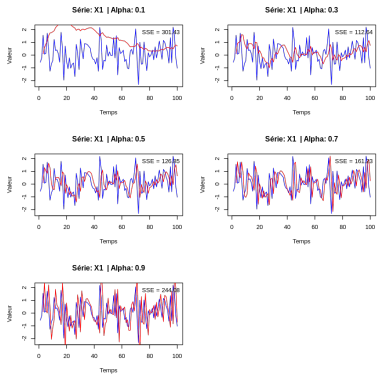

Figure 10: Double exponential smoothing of X_1 for different values of α.

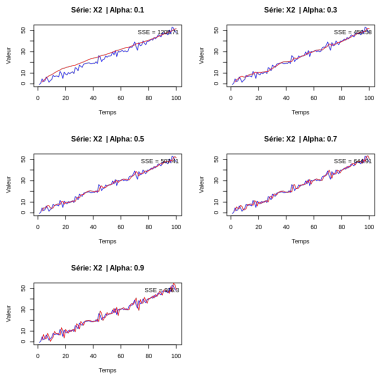

Figure 11: Double exponential smoothing of X_2 for different values of α.

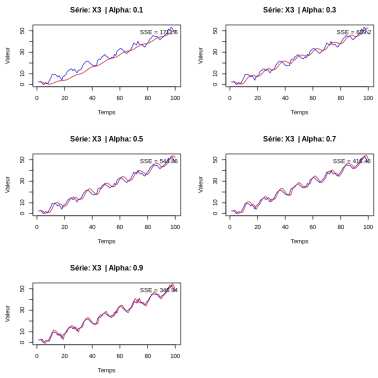

Figure 12: Double exponential smoothing of X_3 for different values of α.

Visually, it appears that for double exponential smoothing, the optimal value $\alpha = 0.9$ also provides the best approximation of the real curves for all three time series. Let us minimize the SSE to determine the optimal values of α for each series.

α	X_1	X_2	X_3
0.1	301.4283	1205.7133	1711.6003
0.3	112.6438	450.5752	639.1996
0.5	126.8529	507.4116	544.8580
0.7	161.2277	644.9107	416.4560
0.9	244.0760	976.3041	346.9360

Table 2: SSE for all series and all values of α. When the optimization criterion is SSE, $\alpha = 0.3$ is optimal for X_1, $\alpha = 0.3$ is optimal for X_2, and $\alpha = 0.9$ is optimal for X_3.

As in (2.1.), we can make the same observation regarding the lag of the smoothing. The SSE for double exponential smoothing is higher than for simple smoothing. Therefore, according to the SSE criterion, simple exponential smoothing is preferable.

2.3. Holt-Winters Exponential Smoothing

Holt-Winters exponential smoothing is applied to X_3 only, as it is the only time series that exhibits a periodic component.

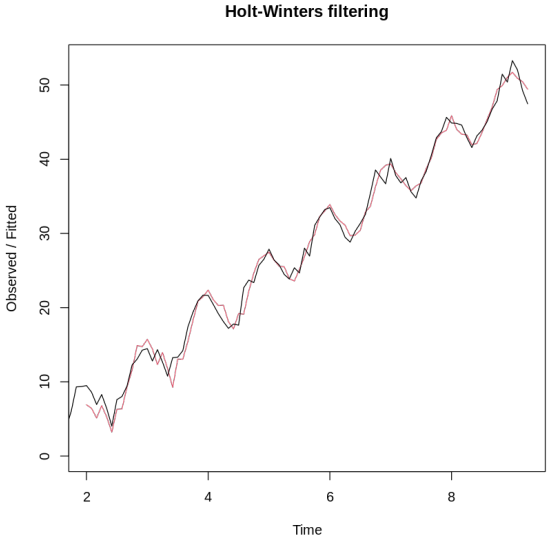

Figure 13: Holt-Winters exponential smoothing with additive seasonal component. $\alpha = 0.1836$, $\beta = 0.0150$, $\gamma = 0.3773$, SSE = 148.1641.

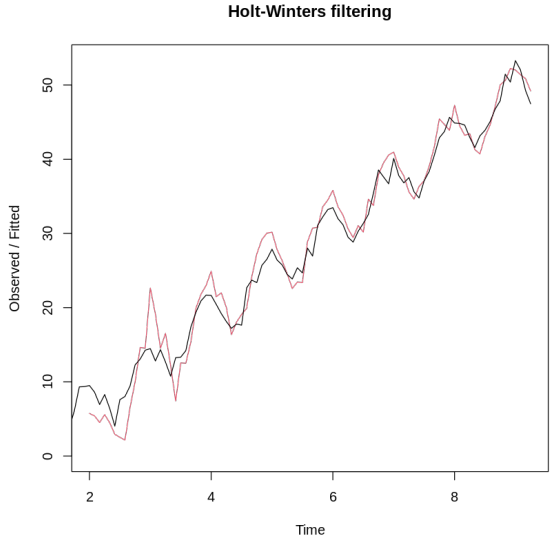

Figure 14: Holt-Winters exponential smoothing with multiplicative seasonal component. $\alpha = 0.0060$, $\beta = 0.0$, $\gamma = 0.8412$, SSE = 486.1205.

To keep this report concise, we did not test the smoothing for different values of the parameters α, β, and γ. Instead, we used the `HoltWinters` function in `R` without specifying these parameters. Once the optimal model was fitted by `R`, we extracted the values of these parameters. According to the SSE criterion, the optimal model for smoothing X_3 among those tested so far is the Holt-Winters exponential smoothing with additive seasonal component, where $\alpha = 0.1836$, $\beta = 0.0150$, $\gamma = 0.$ 148.1641.

3. Lab 2: Smoothing and Forecasting of CO2 Concentration

The time series studied contains CO_2 concentrations near the Mauna Loa volcano in Hawaii from 1959 to 1997.

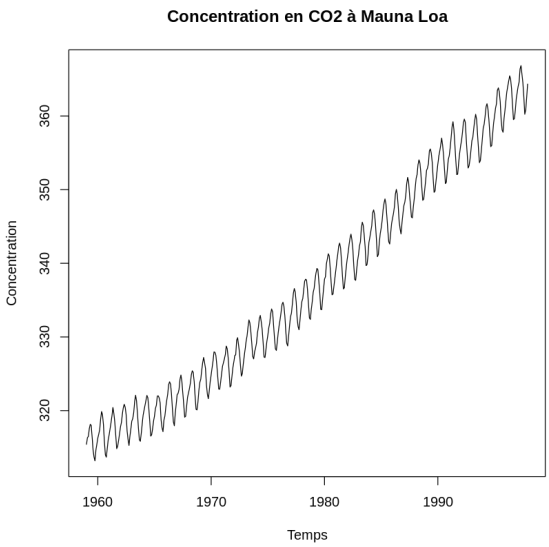

Figure 15: CO_2 concentrations near the Mauna Loa volcano in Hawaii from 1959 to 1997.

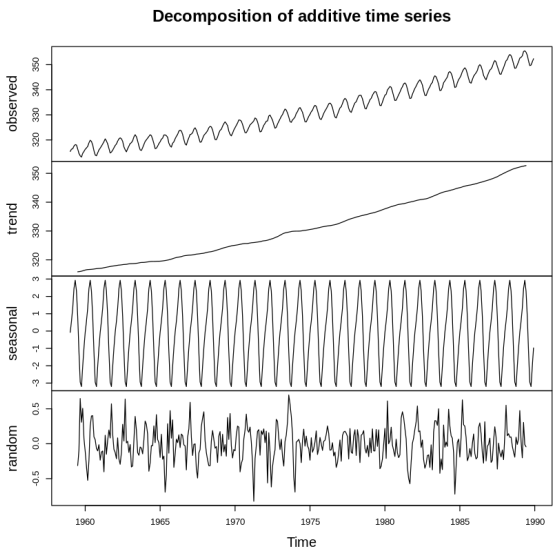

Figure 16: Additive decomposition of CO_2 concentrations. We chose the additive decomposition because the amplitude of the concentration does not change over time.

We split the time series into two parts: a training set covering the years 1959 to 1989, and a test set covering the years 1990 to 1997. The models will be trained on the training set, their performance evaluated on the test set, and the model yielding the best results will be used to forecast the evolution of the time series up to 2007.

As in 2.3., we will apply the smoothing methods in `R` without specifying parameter values. `R` will return the optimal parameters for each model.

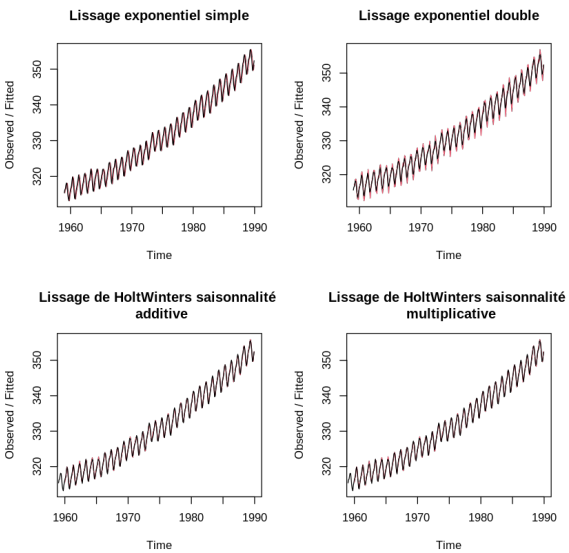

Figure 17: Four exponential smoothing models applied to CO_2 concentrations on the training data.

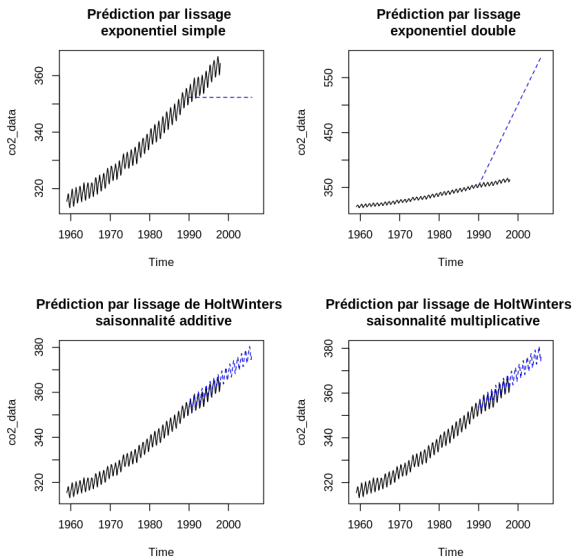

Figure 18: Evaluation of the models on the test set and forecasting of CO_2 concentrations from 1997 to 2007.

Graphically, the smoothing on the training set is satisfactory. However, for forecasting, we would be inclined to choose a Holt-Winters type smoothing: simple and double smoothing produce constant and linear forecasts respectively. Both Holt-Winters models seem viable, but it would be relevant to examine the mean squared errors on the test range.

Model	Mean Squared Error
Simple exponential	54.3214
Double exponential	3813.9202
Holt-Winters additive	1.4358
Holt-Winters multiplicative	1.6235

Table 3: Summary of results obtained.

The Holt-Winters exponential smoothing with additive seasonal component, where $\alpha = 0.4944$, $\beta = 0.0123$, $\gamma = 0.5188$, and MSE $= 1.4358$, is the optimal forecasting model.

4. Lab 2: Smoothing and Forecasting of the CAC40

The time series studied contains the daily closing values of the CAC40 from 1991 to 1998. We will attempt to forecast the closing values of 1998 using exponential smoothing based on the closing values from 1991 to 1997.

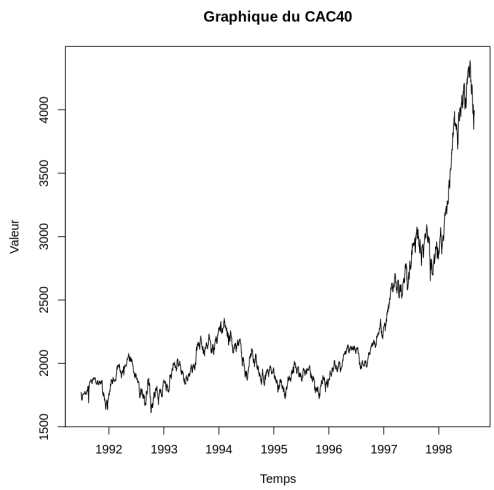

Figure 19: Daily closing values of the CAC40 from 1991 to 1998.

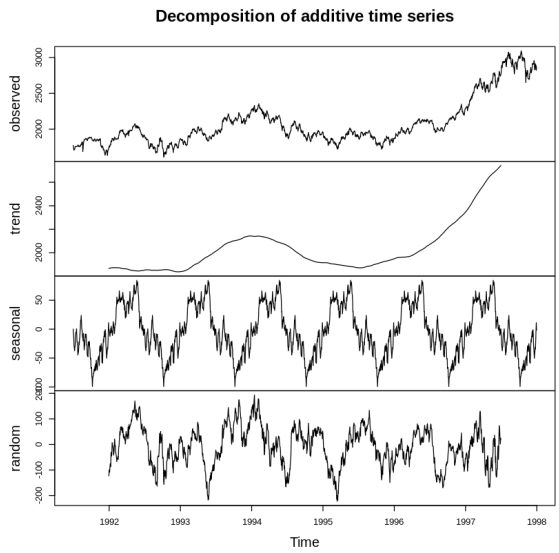

Figure 20: Additive decomposition of the daily closing values of the CAC40.

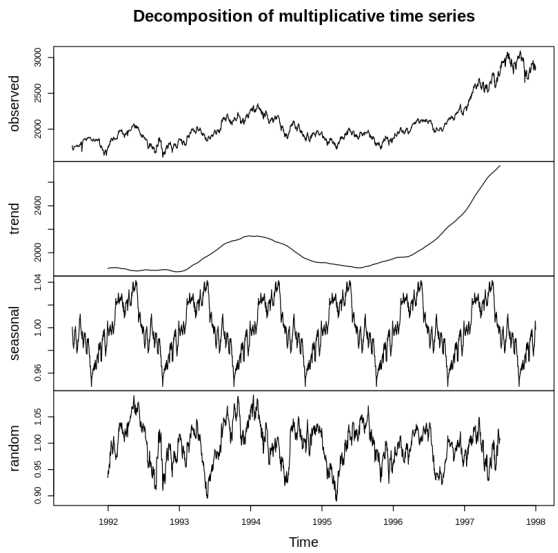

Figure 21: Multiplicative decomposition of the daily closing values of the CAC40.

The time series studied has a periodicity and an upward trend. As in 2.3., we will apply the smoothing methods in R without specifying parameter values. R will return the optimal parameters for each model.

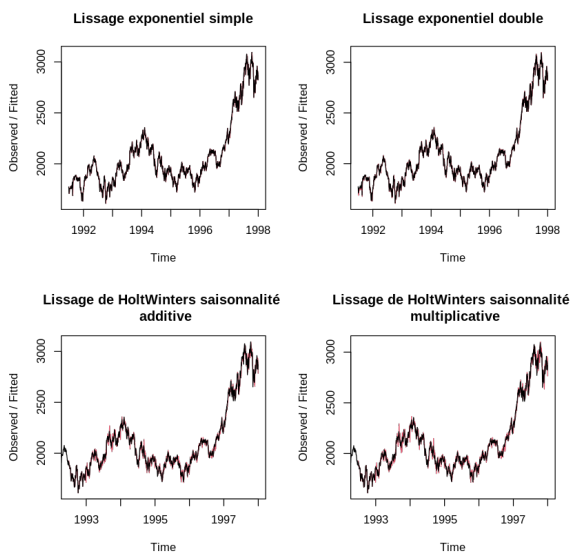

Figure 22: Four exponential smoothing models applied to the daily closing values of the CAC40.

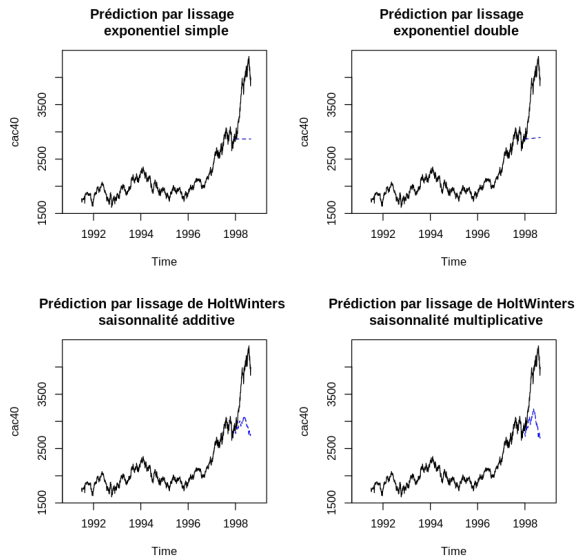

Figure 23: Evaluation of the models on the test set and forecasting of the daily closing values of the CAC40.

Graphically, the smoothing is adequate on the training data. However, on the test set, all four models show errors: simple and double smoothing produce constant and linear forecasts respectively. The Holt-Winters models, meanwhile, follow a pattern similar to a Brownian motion, but do not reflect reality at all.

Model	Mean Squared Error
Simple exponential	926848.5227
Double exponential	899908.0734
Holt-Winters additive	833532.5806
Holt-Winters multiplicative	781013.8203

Table 4: Summary of results obtained.

The Holt-Winters exponential smoothing with multiplicative seasonal component, where $\alpha = 0.8131$, $\beta = 0.0$, $\gamma = 1.0$, and MSE $= 781013.8203$, is the optimal forecasting model.